

Pradeep Karra

United States | +1 (989) 506-3088 | pradeepkarra1@gmail.com | www.linkedin.com/in/karra-pradeep

Professional Summary

I am a results-driven Senior Data Analyst and Data Engineer with over five years of experience in banking, fintech, and higher education. I specialize in building ETL pipelines, data modeling, data cleaning, and data warehousing, with advanced expertise in SQL, Python, Tableau, Power BI, and Excel (VBA, macros, Power Query). I have applied AWS cloud technologies (S3, Redshift, Glue, Lambda, EC2) to manage large-scale data pipelines and automate reporting. I excel at summarizing and interpreting data for key stakeholders such as Subject Matter Experts and Business Partners, ensuring technical solutions directly address business needs. My work has saved hundreds of hours, improved fraud detection accuracy, reduced losses, and empowered leaders to make multimillion-dollar business decisions with real-time dashboards and predictive models. I thrive in Agile teams using Jira, Confluence, and Scrum to manage workflows, prioritize deliverables, and deliver results quickly.

Education

Central Michigan University – Michigan, USA

Aug 2023 - May 2025

Master of Science, Information Technology (GPA: 3.75/4)

St. Martin's Engineering College - Hyderabad, India

Jun 2016 - May 2019

Bachelor of Technology, Electronics & Communication Engineering (GPA: 7/10)

TRR Engineering College – Hyderabad, India

Jun 2013 - May 2016

Diploma, Electrical & Electronics Engineering (GPA: 7.5/10)

Technical Skills

- **MS Office Skills:** Excel (Advanced: Pivot Tables, VLOOKUP/XLOOKUP, INDEX-MATCH, Macros, VBA, Power Query, Data Analysis ToolPak), PowerPoint (Executive Presentations), Word (Documentation), Outlook (Email & Calendar)
- **Languages & Tools:** Python, R, SQL, C, C++, MATLAB
- **Transaction Management:** ETL, Data Wrangling, Post-Trade Processing, Trade Settlement, Apache Airflow, dbt, PySpark
- **Databases:** Snowflake, MySQL, PostgreSQL, SQL Server, Redshift, Teradata
- **Risk Management & Analytics:** Tableau, Power BI, Splunk, Qlik, Matplotlib, Seaborn
- **Cloud Platforms:** AWS (Glue, Redshift, Lambda, S3, EC2), Azure (Data Factory, Synapse), GCP (BigQuery, Dataflow), Databricks, Kafka
- **Compliance & Reporting:** Machine Learning, XGBoost, TensorFlow, Time Series Forecasting, Regulatory Reporting
- **Version Control & Workflow:** GitHub, CI/CD, Jira, Agile, Scrum
- **Collaborative Leadership:** Strong communicator, proven ability to partner with SMEs and Business Partners to align data solutions with business needs

Professional Experience

Central Michigan University | Data Analyst / Engineer

Aug 2023 - May 2025

- Leaders lacked a unified view, so I created Tableau and Power BI dashboards using data from AWS S3 and Snowflake. This cut meeting prep time by 70% and gave directors one trusted view.
- Reporting across ticketing, CRM, and finance systems was inconsistent, so I built ETL pipelines in SQL and Python, cleaned data with Excel Power Query, and stored it in AWS Redshift. This reduced manual reporting by 85% and improved accuracy by 90%.
- NCAA and FERPA compliance reports were manual, so I automated them with SQL, Excel VBA macros, and AWS Lambda scheduling. This saved 200+ hours annually and ensured 100% compliance.
- Forecasts were unreliable, so I built forecasting models in Python deployed on AWS EC2 and combined them with Excel scenario sheets. Accuracy improved by 40% and ROI by 25%.
- KPIs varied across systems, so I standardized metrics using SQL and Excel Power Query with AWS Glue pipelines, improving consistency by 95%.
- Dashboards refreshed only once per day, so I optimized SQL queries and leveraged S3 data refreshes to update every 15 minutes, enabling live monitoring.
- Worked in Agile sprints with Jira, collaborating with SMEs, Finance, and IT stakeholders to summarize findings and align data reporting with executive decision-making.

Genpact | Senior Data Analyst / Engineer (Google Pay Project)

Jun 2021 - Jul 2023

- Fraud detection was slow, so I developed SQL and Python anomaly checks on 500GB+ daily transactions processed via AWS Glue. Fraud was flagged in seconds instead of hours.
- Refund complaints piled up, so I created Tableau dashboards and Excel macros using refund and complaint data stored in AWS Redshift. Resolution time dropped by 40% and trust improved.
- Monitoring was delayed, so I integrated Kafka real-time streams into Snowflake, staging files in AWS S3 and generating alerts in Excel reports. Detection accuracy improved by 50%.
- Compliance reports were error-prone, so I automated them in SQL, Python, and Excel VBA with validation through AWS Lambda. Preparation time was cut by 60%.
- Cashback workflows caused repeated complaints. Using SQL and Excel pivot reports on AWS Redshift data, I redesigned workflows with engineering and business partners. Complaints dropped by 30%.
- Data quality slowed reporting, so I built Python cleaning scripts and Excel validation macros for S3 data. Reliability improved by 35%.
- Partnered with SMEs and Business Partners to present findings from dashboards in executive reviews, ensuring business concerns were translated into data-driven actions. Managed sprint deliverables in Jira and documented workflows in Confluence.

ICICI Bank | Analyst, Risk & Fraud Analytics

Jul 2019 - Jul 2021

- Fraud reporting took 3 days, so I automated it with SQL stored procedures, Excel VBA macros, and AWS Glue pipelines, reducing reporting time to 11 hours.
- Fraud investigations were backlogged, so I analyzed 10M+ daily transactions in SQL, Python, and Excel, migrating fraud data into AWS S3 for scalable storage. Backlog was reduced by 35%.

- False positives drained resources, so I built Power BI dashboards and Excel risk models pulling data from AWS Redshift, which reduced false alerts by 25%.
- High-risk accounts weren't prioritized, so I developed a fraud scoring model in SQL and Python hosted on AWS EC2, cutting resolution time by 30%.
- ATM and online fraud patterns were missed, so I ran time-series analysis in Python and Excel on logs stored in S3 buckets, leading to RBI rules that reduced ATM fraud by 15%.
- Reports risked audit failures, so I standardized fraud reporting with SQL, Power BI, and AWS Lambda jobs, ensuring RBI compliance and avoiding penalties.
- Coordinated tasks in Agile sprints using Jira, summarizing fraud analytics insights for SMEs, Compliance teams, and regulators to support policy changes.

Lampex Electronics Limited | PCB Manufacturing Intern

May 2015 - Nov 2015

- Soldering defects slowed production, so I practiced wave, reflow, and silver soldering while documenting results in Excel, improving assembly quality.
- LED PCBs needed thermal control, so I worked with MCPCB technologies and tracked results in Excel, improving product durability.
- Supported PCB etching, lamination, drilling, and plating, logging metrics in Excel to give supervisors better visibility.
- Paper-based reporting slowed reviews, so I built Excel pivot dashboards summarizing defects and yields, supporting faster quality reviews.
- Learned solder mask application, conformal coating, and rework, while documenting tasks with Excel checklists that improved process consistency.
- Collaborated with production engineers and SMEs to interpret Excel reports and suggest corrective actions, improving overall quality outcomes.
- PCB inspections lacked structured reporting, so I logged defects in Excel and automated checks with macros, improving accuracy and reducing rework.

Projects

Sentiment Analysis using R

- Collected and cleaned social media and review data, applied sentiment models in R, and visualized trends in ggplot2.
- Helped businesses interpret customer sentiment with SMEs and adjust campaigns.

Leaf Disease Detection using MATLAB and Python

- Built classification models using MATLAB preprocessing and Python TensorFlow.
- Detected crop diseases early, helping farmers and researchers reduce losses.

WeaveTrack Inventory Management System

- Developed a Python (Tkinter UI) + Excel VBA app to manage suppliers, stock, and purchase orders.
- Automated ETL reduced errors by 80%.

Fraud Detection Analytics

- Combined KYC and transaction data in SQL, applied anomaly detection in Python, and built Power BI/Excel dashboards.
- Summarized fraud insights for compliance SMEs, reducing false positives by 25%.

Google Pay Cashback Monitoring

- Processed refund/complaint data in SQL, validated with Excel macros, and visualized in Tableau.
- Improved visibility and reduced complaint resolution time by 40%.

Athletics Executive Dashboard

- Automated ETL in SQL/Python to integrate CRM, ticketing, and finance data into Tableau and Power BI dashboards.
- Summarized KPIs for directors and business partners, improving multimillion-dollar planning.

Certifications

- **AWS Cloud Practitioner:** LinkedIn Learning
- **Analyzing Data in Looker, LookML Modeling, Vertex AI Prompt Design:** Google Cloud
- **Lean Six Sigma Green Belt (Process Improvement & Risk Management):** Central Michigan University
- **Deloitte Data Analytics Simulation:** Forage
- **Citi Markets Quantitative Analysis (MQA):** Forage